

Geometry
Unit 13
Lesson 12 Practice

Name Answer Key
Date _____
Period _____

Right triangle MTH has vertices at $M(3, -8), T(8, 4), H(8, -8)$. Right triangle DPT has vertices at $D(-3, -4), P(-5, -6), T(-3, -6)$.

1. Determine if $\triangle MTH$ and $\triangle DPT$ are congruent, similar, or neither. **Neither**

$$MT = \sqrt{(8-3)^2 + (4+8)^2} = 13$$

$$DP = \sqrt{(-5+3)^2 + (-6+4)^2} = \sqrt{8} = 2\sqrt{2}$$

$$TH = \sqrt{(8-8)^2 + (4+8)^2} = 12$$

$$PT = \sqrt{(-3+5)^2 + 0^2} = \sqrt{4} = 2$$

$$HM = \sqrt{(8-3)^2 + 0^2} = 5$$

$$TD = \sqrt{(-6+4)^2 + (0)^2} = \sqrt{4} = 2$$

2. What definition, theorem, or property justifies your conclusions?

Does not satisfy SSS congruence or similarity.

3. Describe the transformations that would show that the figures are congruent or that they are similar. State which image you used as the pre-image.

Using triangle MTH as the pre-image, shift 11 units left and 2 units up. This will align the right angles, and you can see the triangles are neither congruent or similar.

Determine if the given polygons are congruent, similar, or neither.

4. Trapezoid $TRPS$: $T(1,1), R(4,5), P(7,5), S(10,1)$
Trapezoid $RTLF$: $R(3,3), T(12,15), L(21,15), F(30,3)$

- ☐ Congruent
☒ Similar
☐ Neither

$$TR: \sqrt{3^2 + 4^2} = 5$$

$$RT: \sqrt{9^2 + 12^2} = 15$$

$$RP: \sqrt{0^2 + 3^2} = 3$$

$$TL: \sqrt{9^2} = 9$$

$$PS: \sqrt{3^2 + 4^2} = 5$$

$$LF: \sqrt{9^2 + 12^2} = 15$$

$$TS: \sqrt{9^2 + 0^2} = 9$$

$$RF: \sqrt{27^2} = 27$$

5. Kite $GHKM$: $G(0,0), H(1,-4), K(3,-3), M(4,-1)$.
Kite $RSTV$: $R(0,0), S(2,-8), T(6,-6), V(8,-2)$.

- ☐ Congruent
☒ Similar
☐ Neither

$$GH: \sqrt{1^2 + 16} = \sqrt{17}$$

$$RS: \sqrt{2^2 + 8^2} = \sqrt{68} = 2\sqrt{17}$$

$$HK: \sqrt{4 + 1} = \sqrt{5}$$

$$ST: \sqrt{16 + 4} = \sqrt{20} = 2\sqrt{5}$$

2:1 ratio

6. Rhombus $DEFG$: $D(6,4), E(-1,0), F(-2,-8), G(5,-4)$
Rhombus $BKMH$: $B(3,2), K(-0.5,0), M(-1,-4), H(2.5,-2)$

- ☐ Congruent
☒ Similar
☐ Neither

Corresponding angles are congruent. Since sides are congruent, ratios of corresponding sides are congruent.

Lane and Tonymy are determining if parallelograms $HMTY$ and $NRPX$ are congruent, similar, or neither. Parallelogram $HMTY$ has vertices at $H(5,6)$, $M(-3,3)$, $T(-4,-1)$, and $Y(4,2)$. Parallelogram $NRPX$ has vertices at $N(10,12)$, $R(-6,6)$, $P(-8,-2)$, and $X(8,4)$. Their work is shown.

Lane	Tonymy
Parallelograms $HMTY$ and $NRPX$ are similar because $\frac{MH}{RN} = \frac{XR}{PN}$	Parallelograms $HMTY$ and $NRPX$ are not similar because $\frac{MH}{RN} = \frac{XN}{PR}$

7. Who do you agree with?

Neither, Lane used diagonals instead of sides, and Tonymy did not setup the proportion correctly.

8. Explain the mistake that was made and how to correct it.

$$\frac{HM}{RN} = \frac{\sqrt{8^2 + 3^2}}{\sqrt{16^2 + 6^2}} = \frac{\sqrt{73}}{\sqrt{292}}$$

$$\frac{YH}{XN} = \frac{\sqrt{1^2 + 4^2}}{\sqrt{2^2 + 8^2}} = \frac{\sqrt{17}}{\sqrt{68}}$$

$$\frac{\sqrt{73}}{\sqrt{292}} = \frac{\sqrt{17}}{\sqrt{68}}$$

Explanations may vary but should include that the parallelogram are similar due to having equivalent ratios for the corresponding sides.